

Research Methods

Day 11 · Session 11: Interpreting Findings & Discussion

Outline for Today

- 1. Draft Report Feedback**
- 2. How to Write a Discussion Section: The Four-Move Structure**
- 3. Team Activity: Discussion Synthesis**
- 4. Synthesizing Your Findings with the Literature**
- 5. Presentations: Discussion + Conclusion Slides**
- 6. Scientific Storytelling: Presentation Transitions**



Session 11: Interpreting Findings & Discussion

Learning Objectives

Connect findings to the question & literature

Weave your data findings into a coherent narrative that ties back to the research question and to what others have found.

Write a Discussion that contextualizes without overclaiming

Interpret your results honestly — say what they mean, and no more than the data can support.

Identify limitations honestly and constructively

Name the caveats that most threaten your interpretation — and what they imply.

Begin integrating all sections into a complete presentation

Assemble intro, methods, results, discussion, and conclusion into one connected story.

From Results to Meaning

Results tell the audience *what* you found. The Discussion tells them *what it means*.

The Results section reports the numbers. The Discussion is where you become the interpreter of your own evidence.

A strong Discussion answers four questions:

- **So what?** — why should anyone care?
- **Why?** — what might explain the finding?
- **Compared to what?** — how does it sit against the literature?
- **What are the caveats?** — where could you be wrong?

The interpreter's job

You are the person who knows this dataset best. The Discussion is your chance to explain what the reader should take away — and to be honest about the limits of that claim.

Discussion ≠ repeat of Results

Don't just restate every number again. Pick the findings that matter and interpret them.



Morning: ResearchCORE

Draft Report Feedback (20 min)

Marked-up drafts returned

Each RcS team gets their draft back with comments in the margins.

Two class-wide teaching points

We'll highlight two issues that showed up across *many* drafts — the patterns worth everyone hearing.

10 minutes to review

Teams read their feedback together and decide what to revise first.

As you read your feedback, ask:

- Which comment, if fixed, would improve the paper the *most*?
- Is any comment pointing at a problem that appears in more than one place?
- What can we fix *today* while we draft the Discussion?

10:00



How to Write a Discussion Section

The Four-Move Structure

Every paragraph of a good Discussion moves through the same four steps:



1. Restate

Say the finding in plain language — no jargon, no table numbers.



2. Explain

Why might this be happening? What is the mechanism?



3. Compare

Agree, disagree, or add nuance to what the literature said?

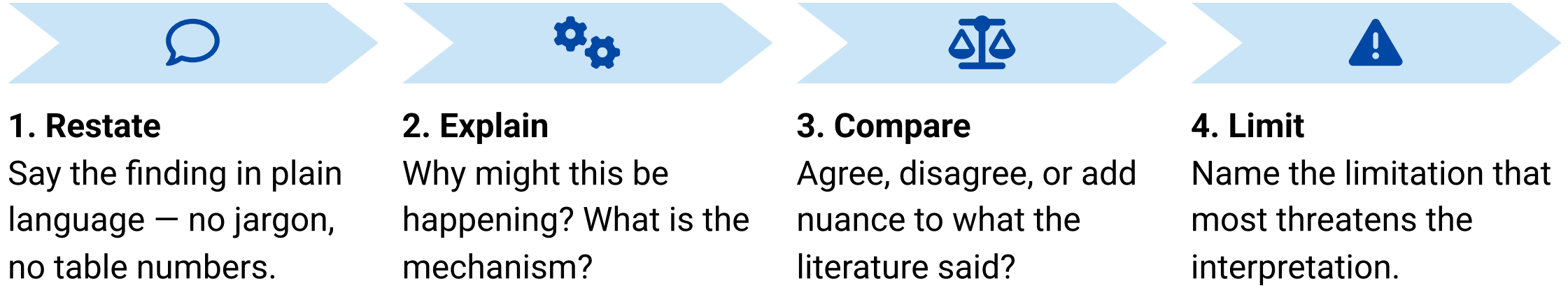


4. Limit

Name the limitation that most threatens the interpretation.

The Four-Move Structure

Every paragraph of a good Discussion moves through the same four steps:



Use it as a template. Take one key finding, walk it through all four moves, and you have a complete Discussion paragraph.

Move 1 – Restate the Finding in Plain Language

Plain language, not table-speak

Translate the statistic into a sentence a non-expert would understand.

- Drop the p -values and coefficients for a moment.
- Say what actually happened to whom.
- Anchor it to your research question.

Instead of:

"The coefficient on treatment was 4.2 ($p < 0.05$)."

Say:

"Students who used the study app scored about 4 points higher on the final test than those who did not."

Move 2 – Explain It

Why might you have found this? Offer a *mechanism* – a plausible story for how the cause produces the effect.

- What do we already know about *why* this relationship could exist?
- Is there a chain of steps linking your variables?
- Could something else be driving it? (name the **alternative explanation**)

Example mechanism

"The app may help because it spaces practice over time, and spaced repetition is known to improve retention."

Alternative explanation

"But motivated students may have both chosen the app and studied more – so we can't rule out self-selection."

Move 3 – Compare to the Literature

Does your finding **agree**, **disagree**, or **add nuance** to what was already known?



Agrees

"This is consistent with Smith (2021), who also found a positive effect of spaced practice."



Disagrees

"In contrast to Lee (2020), we found no gender difference – possibly because our sample was older."



Adds nuance

"Building on Gomez (2022), we show the effect holds, but only for students who used the app weekly."

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Adds nuance

"Building on Gomez (2022), we show the effect holds, but only for students who used the app weekly."

Every comparison needs a citation. Tie each finding to at least one source from your literature review.

Move 4 – Identify the Key Limitation

Every study has limits. **Name the one that most threatens your interpretation** — not a laundry list.

Ask:

- What would a skeptic attack first?
- Does this limitation change *what the finding means*?
- What would you do differently with more time or data?

A limitation with teeth

"Because students chose whether to use the app, our estimate may reflect who they are, not what the app does. A randomized trial would settle this."

Honest ≠ apologetic

State the limit, say what it implies, and move on. Don't undermine your whole study.

Interpret Without Overclaiming

✗ Overclaiming

- *"This proves the app causes higher scores."*
- *"Our results show this will work everywhere."*
- Treating correlation as causation.
- Generalizing beyond your sample.

✓ Calibrated claims

- *"Our results **suggest** the app **is associated with** higher scores."*
- *"**Within our sample**, students who..."*
- Hedging language: *may, suggests, is consistent with.*
- Scope conditions: for whom, where, when.

The test: could you defend every sentence of your Discussion if the author you cited were sitting in the front row?



Team Activity: Discussion Synthesis

Team Activity – Discussion Synthesis (30 min)

Draft your Discussion

Take your **main finding** and write one paragraph using the **four-move structure**: restate → explain → compare → limit.

Bring in the literature

Connect the finding to at least **one source** from your literature review.

Instructor circulates

Be ready for probing questions.

Probing questions to expect:

- *"Why do you think you found that?"*
- *"What alternative explanation can you think of?"*
- *"Who would challenge this finding – and how would you respond?"*

30:00

Mission Sheet: Synthesizing Findings

Before you leave the activity, your team should be able to fill in **four lines**:

1 **Main conclusion** — in one plain-language sentence, what did you find?

2 **Comparison to literature** — one source your finding agrees with, disagrees with, or extends (with citation).

3 **One important limitation** — the caveat that most threatens your interpretation.

4 **One next step** — what should the next study do to build on or fix this?

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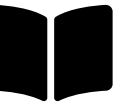
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Then: share your four lines with a neighboring team and get quick feedback before the next segment.



Synthesizing Your Findings with the Literature

What Does Synthesis Mean?

Synthesis is more than summary. It means combining ideas from different sources to see *how they relate* — not just what each one says.

A good discussion doesn't list studies one after another. It **weaves** them together around *your* finding.

Ask of your sources:

- Do the authors **agree** or **disagree**?
- Are they studying the same issue from **different angles**?
- Does one source **support, expand on, or challenge** another?

The "discuss" checklist

To *discuss* a result means to:

1. Restate the key finding
2. Compare to what the literature said
3. Offer an explanation for it
4. Acknowledge limitations
5. Suggest what comes next

Group Ideas, Not Authors

✗ Don't organize by author:

"Smith says... then Jones says... then Lee says..."

This reads like a list, not an argument.

✓ Organize by theme, issue, or trend:

"Many studies agree that social media affects attention span (Smith, 2021; Lee, 2020), but a few challenge this idea (Gomez, 2022)."

One sentence, three sources, one clear point.

Use Comparative Language

Helpful connecting phrases

- *"Similarly..."*
- *"In contrast..."*
- *"Building on..."*
- *"While some researchers argue..., others suggest..."*
- *"There is a general consensus that..."*

Highlight gaps or conflicts

Point out where sources disagree or where more research is needed.

"While most studies link sleep to better memory, few explore how this differs by age group."

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"While most studies link sleep to better memory, few explore how this differs by age group."

End with the big picture: what have you learned from all the sources *combined*? What patterns, debates, or gaps set the stage for your own finding?



Presentations: Discussion + Conclusion Slides

Present the Ending (30 min)

Each team presents the **full ending** of their presentation – the Discussion and Conclusion slides.

Feedback focuses on two questions:

↔ **Does the discussion connect back?**

Do the findings tie clearly to the **original research question?**

🚩 **Does the conclusion land?**

Does it leave the audience with **one clear takeaway?**

Draft your Discussion + Conclusion slides around three questions:

- What do our results **mean**?
- Did they **support or challenge** our hypothesis?
- **Who should care** about this?

30:00

Anatomy of a Strong Conclusion Slide

One takeaway

Lead with the single sentence you want the audience to remember.

Answers the question

Explicitly close the loop back to the question you opened with.

Honest caveat

One limitation, stated plainly — it builds credibility.

Where next

A forward-looking line: what this opens up or what should follow.

Avoid ending on a wall of numbers or a plain "Thank you." End on *meaning*.



Scientific Storytelling: Presentation Transitions

Verbal Transitions Between Sections (10 min)

Smooth transitions signal a **well-told story**. Practice the sentences that carry your audience from one section to the next.

→ **Methods** → **Results**: *"Now that we've seen the methods, let me walk you through what we found..."*

→ **Results** → **Discussion**: *"Our results suggest that... This aligns with / differs from what [Author, Year] found in their study..."*

→ **Discussion** → **Conclusion**: *"So what does all of this mean? Here is the one thing we want you to remember..."*

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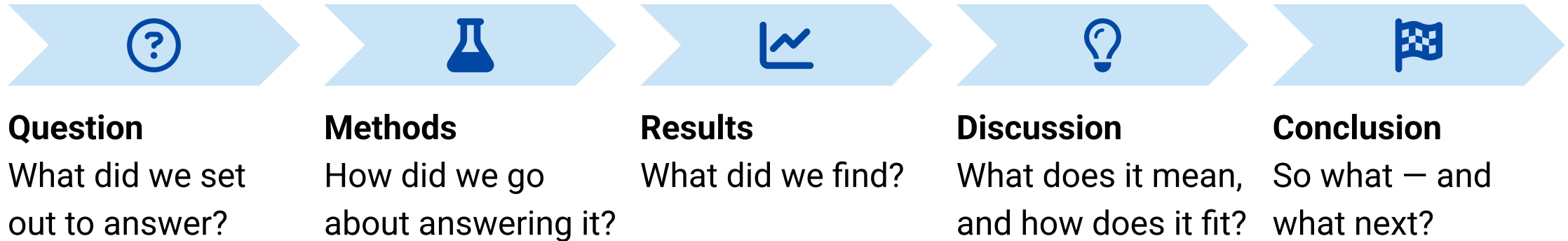
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→ **Discussion** → **Conclusion**: *"So what does all of this mean? Here is the one thing we want you to remember..."*

Practice out loud. The transition is where a presentation feels either seamless or stitched together. Rehearse the hand-offs, especially when the speaker changes.

10:00

Putting the Whole Story Together



Integration is today's goal: every section should point back to the question and forward to the takeaway. That through-line *is* the story.



Homework & Teamwork

Homework – Teamwork

As a team

- Write your **synthesis**: conclusion sentence, comparison to one literature source, one limitation, one next step.
- Draft the **Discussion slide** – *"What do our results mean?"* – connecting each finding to at least one source from your literature review.
- Finalize slide content and **assign who presents which slide**. Then **practice!**

Slide check (async is fine)

- Are all charts labeled? Does the story flow? Identify **3 specific things to fix** in tomorrow's session.

Individual (evening)

- Draft your **Discussion paragraph**: did the results support your hypothesis? How do findings compare to the literature? *(200–250 words)*

Written report (RcS)

- Add a **Discussion / Conclusion** section to your written report. Include at least **2 literature comparisons with citations**. *(Target: 300–400 words)*

Session 11: What You Can Do Now

Interpreting findings:

- Take a key result and move it through the **four-move structure** — restate in plain language, explain the mechanism, compare to the literature, and name the limitation that most threatens it.

Writing the Discussion:

- Contextualize results **without overclaiming** — use calibrated language (*suggests, is associated with*), state scope conditions, and treat correlation as correlation.
- **Synthesize** sources by theme rather than by author, using comparative language to show agreement, conflict, and gaps.

Building the presentation:

- Draft **Discussion and Conclusion slides** that connect findings back to the research question and leave one clear takeaway.
- Deliver smooth **verbal transitions** between sections so the whole talk reads as one coherent story.